

DATA ANALYTICS PROJECT SUMMARY

Superstore Sales & Profit Analysis

An end-to-end analytics pipeline covering data cleaning, exploratory analysis, SQL-based querying, and an interactive Power BI dashboard — built on 9,994 real US retail orders spanning 2014–2017.

Python

Pandas, Jupyter

SQL

SQLite & SQL Server

BI

Power BI Desktop

1. Project Overview

This project takes a raw Excel export of Superstore order data through a complete analytics workflow: cleaning and feature engineering in Python, structured querying in SQL (first SQLite, then a live SQL Server database), and a final 3-page interactive Power BI dashboard built on top of that data.



2. Project Files

.XLS	Sample_Superstore.xls Original raw dataset – 9,994 orders, 21 columns (Order/Ship dates, Customer, Product, Sales, Profit, Discount, etc.), 2014–2017.
.CSV	Superstore_Cleaned.csv Cleaned dataset with 6 engineered features added: Order Year, Order Month, Order Month-Year, Shipping Days, Profit Margin – 27 columns total, zero nulls or duplicates.
.IPYNB	Superstore_Analysis.ipynb Jupyter notebook – data cleaning, exploratory analysis with 8 chart types, SQLite querying, and the SQL Server upload connection (pyodbc/SQLAlchemy).
.DB	superstore.db SQLite database generated from the notebook, containing the full cleaned dataset in a single `orders` table (9,994 rows) for local SQL querying.
.SQL	SQLQuery1.sql T-SQL query library run against SQL Server (`dbo.customer` table) – category profitability, YoY growth, loss-making sub-categories, top customers, monthly running totals, and more.
.PBIX	Superstore_Performance_Dashboard.pbix Final Power BI dashboard – 3 pages (Overview, Profitability, Customers & Operations), 37 visuals total, connected live to SQL Server with synced slicers.

3. Dashboard — Executive Overview

Page 1 of the Power BI dashboard: top-line KPIs, the monthly sales/profit trend, category and segment breakdowns, and regional sales.

Superstore Performance Dashboard

US Retail Orders · 2014-2017 · Sales, Profit & Customer Analytics

Executive overview

Profitability

Customers & operations

YEAR

All years

REGION

All regions

CATEGORY

All categories

TOTAL SALES

\$2.30M

TOTAL PROFIT

\$286.4K

PROFIT MARGIN

12.5%

TOTAL ORDERS

5,009

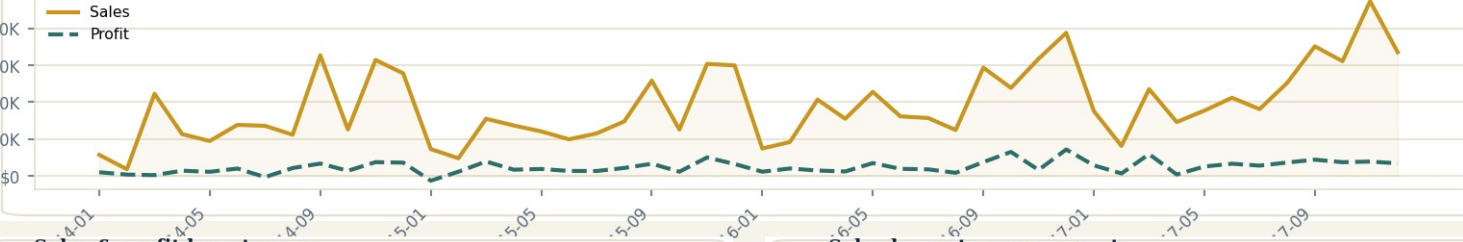
AVG DISCOUNT

15.6%

CUSTOMERS

793

Monthly sales & profit trend



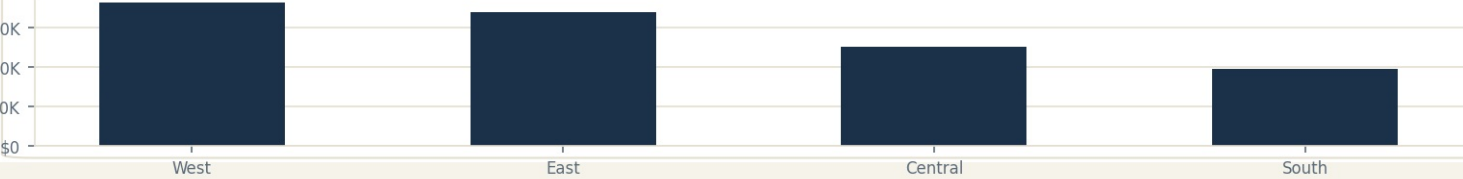
Sales & profit by category



Sales by customer segment



Sales by region



4. Dashboard — Profitability

Page 2: where the business is losing money — sub-category profit, discount-vs-profit relationship, and a ranked loss table.

Superstore Performance Dashboard

US Retail Orders · 2014-2017 · Sales, Profit & Customer Analytics

Executive overview

Profitability

Customers & operations

YEAR

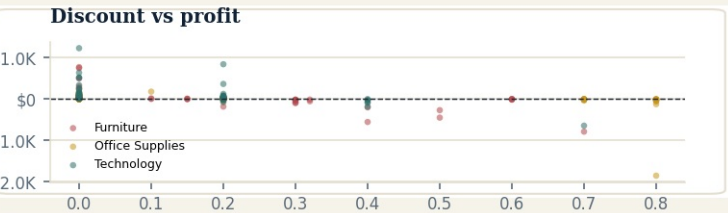
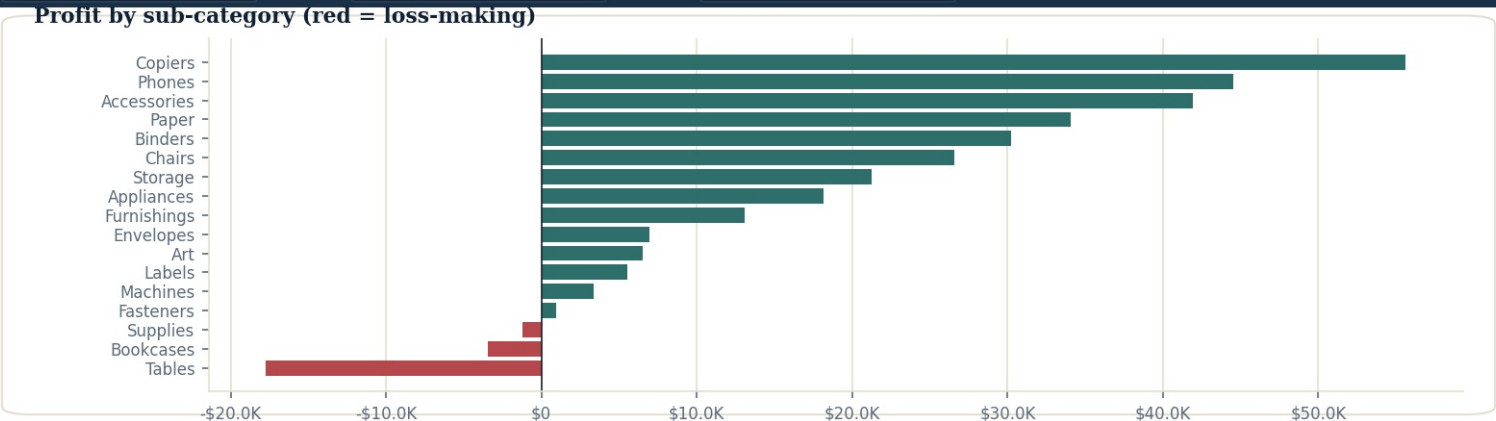
All years

REGION

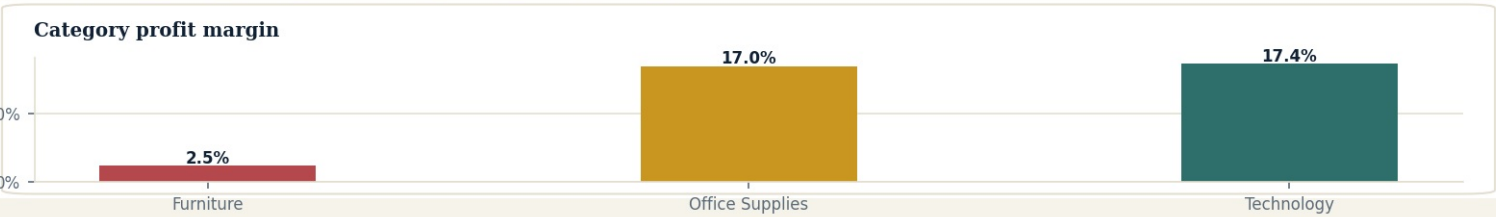
All regions

CATEGORY

All categories



Loss-making sub-categories	
SUB-CATEGORY	PROFIT
Tables	-\$17.7K
Bookcases	-\$3.5K
Supplies	-\$1.2K



5. Dashboard — Customers & Operations

Page 3: top customers by lifetime sales and shipping speed/volume by ship mode.

Superstore Performance Dashboard

US Retail Orders · 2014-2017 · Sales, Profit & Customer Analytics

Executive overview

Profitability

Customers & operations

YEAR

All years

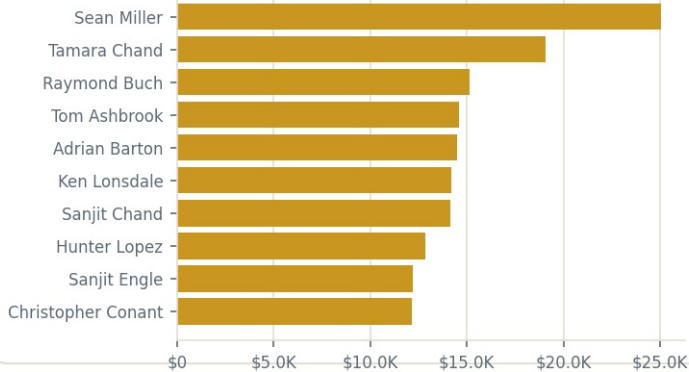
REGION

All regions

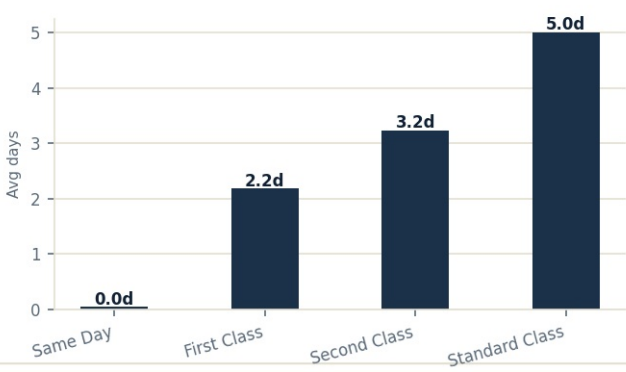
CATEGORY

All categories

Top 10 customers by sales



Shipping performance by mode



Ship mode order volume

SHIP MODE	ORDERS	AVG SHIPPING DAYS
Same Day	264	0.04
First Class	787	2.18
Second Class	964	3.24
Standard Class	2,994	5.01

6. Key Metrics

TOTAL SALES \$2.30M	TOTAL PROFIT \$286.4K	PROFIT MARGIN 12.47%
TOTAL ORDERS 5,009	UNIQUE CUSTOMERS 793	DATE RANGE 2014-2017

Sales & Profit by Category

CATEGORY	SALES	PROFIT
Technology	\$836.2K	\$145.5K
Furniture	\$742.0K	\$18.5K
Office Supplies	\$719.0K	\$122.5K

Sales & Profit by Region

REGION	SALES	PROFIT
West	\$725.5K	\$108.4K
East	\$678.8K	\$91.5K
Central	\$501.2K	\$39.7K
South	\$391.7K	\$46.7K

Sales & Profit by Year

YEAR	SALES	PROFIT
2014	\$484.2K	\$49.5K
2015	\$470.5K	\$61.6K
2016	\$609.2K	\$81.8K
2017	\$733.2K	\$93.4K

Weakest Sub-Categories (Profit)

SUB-CATEGORY	PROFIT
Tables	-\$17.7K
Bookcases	-\$3.5K
Supplies	-\$1.2K

7. Methodology

Python — Cleaning & Exploratory Analysis

Loaded the raw `.xls` file, verified there were no missing values or duplicate rows, converted date columns to proper datetime types, and engineered 6 new analysis-ready fields (Order Year, Order Month, Shipping Days, Profit Margin, etc.). Explored the data across 8 dimensions: monthly sales/profit trend, category and sub-category profitability, regional performance, discount-vs-profit correlation, top customers, and shipping mode performance.

SQL — Structured Querying

The cleaned dataset was first loaded into a local `SQLite` database directly from the notebook for quick validation, then uploaded to a live `SQL Server` instance via `pyodbc` + `SQLAlchemy` for a production-style setup. A 10-query library answers core business questions: category profitability, year-over-year growth, loss-making region/sub-category combinations, top customers by lifetime sales, and more — using GROUP BY, HAVING, and window functions (running totals).

Power BI — Interactive Dashboard

Connected live to the SQL Server table, built a `DateTable` for time-intelligence, and wrote 7 DAX measures (Total Sales, Total Profit, Profit Margin %, Avg Discount, Avg Order Value, Avg Shipping Days, Total Orders). The dashboard spans 3 pages with synced slicers (Year, Region, Category) so filtering on one page updates all three:

PAGE	PURPOSE	KEY VISUALS
Overview	Executive summary	4 KPI cards, monthly trend line, category bar chart, segment donut, region bar chart
Profitability	Where money is lost	Sub-category profit bars (red/teal), discount-vs-profit scatter, loss-making table
Customers & Operations	Who buys, how it ships	Top 10 customers, shipping speed by mode, ship mode summary table

8. Key Insights

Furniture underperforms on margin. It generates the 2nd-highest sales (\$742K) but by far the lowest profit (\$18.5K, ~2.5% margin) — Technology and Office Supplies both convert sales to profit far more efficiently.

Tables are a consistent loss-maker. The single worst sub-category at −\$17.7K profit, driving most of Furniture's weak margin — a strong candidate for repricing or discount-policy review.

The West region leads on both fronts. Highest sales (\$725K) and highest profit (\$108K) — a benchmark for what's working elsewhere.

Sales grew every year from \$484K (2014) to \$733K (2017), with profit growth following a similar upward trend.